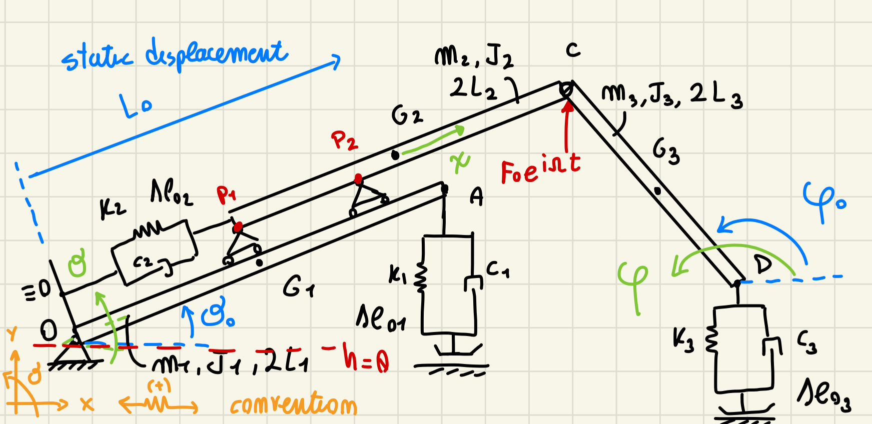
ESE 13 PREPARATION FOR PROBLEM 1 (EXAM)

3 Rigid bodies



1) • Linearized eq of motion
2) • X, ω
3) • FRF with output y_C, X_D
4) • FRF of H_0 respect $F_0 e^{i\Omega t}$
5) • $\Delta l_{01}, \Delta l_{02}, \Delta l_{03}$? IF change configuration

▶ DOF ? $3 \times 3 = 9$ DOF

HINGE (O) − 2 DOC

HINGE (C) − 2 DOC → 3 DOF

CARTS (P_1, P_2) − 2 DOC

to choose 3 indip cohordinates:
- θ respect horizontal axis $= \theta_0 + \theta$
- x: relative displacement of BAR 2 respect 1 measured respect G_2 (dynamic component of displacement)
- φ angular rotation of III BAR

$$\underline{x} = \begin{bmatrix} \theta \\ x \\ \varphi \end{bmatrix} \qquad \underline{x}_0 = \begin{bmatrix} \theta_0 \\ 0 \\ \varphi_0 \end{bmatrix}$$

ENERGETIC QUANTITIES

$$T = \frac{1}{2} m_1 V_{G_1}^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{G_2}^2 + \frac{1}{2} m_2 \dot{y}_{G_2}^2 + \frac{1}{2} J_2 \dot{\theta}^2 + \frac{1}{2} m_3 \dot{x}_{G_3}^2 + \frac{1}{2} m_3 \dot{y}_{G_3}^2 + \frac{1}{2} J_3 \dot{\varphi}^2$$

G_1 Translation

body 2 Rotate as the FIRST

considering that G_1 is connected to the ground by HINGE, easy absolute traj/vel as $L_1 \dot{\theta} = |V_{G_1}|$

we don't know absolute traj and vel of G_2 ⇒ we should write $V_{G_2 x/y}$

cart P_1/P_2 allow relative translation BUT NOT rotation!
BAR 1/2 rotate same way

$$D = \frac{1}{2} \sum_{i=1}^{3} c_i \Delta \dot{L}_i^2 \qquad V_{EL} = \frac{1}{2} \sum_{i=1}^{3} k_i \Delta L_i^2 \qquad V_g = \sum_{i=1}^{3} m_i g h_{G_i}$$

vertical position of G_i respect reference plane

$h_{G_i} > 0$ over $h = 0$ ref

$\delta W = F_0 e^{i\Omega t} \delta y_c$ (scalar product) solved

JUST F_0 acting on C vertically (so related to y_c)

$[M_{pn}]$, $[K_{pn}]$, $[C_{pn}]$ as usual

$\Rightarrow$ than compute LINEARIZED displacement $x_{G_2}, y_{G_2}, x_{G_3}, y_{G_3}, \Delta l_i$ $i=1,2,3$, y_c, h_{G_i} $i=1,2,3$

from here: KINEMATIC relationship

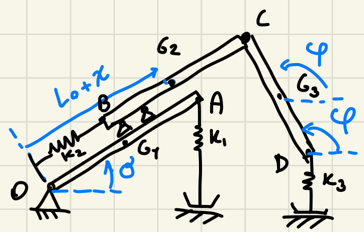
1) super-imposition approach
(apply one effect at a time) $\rightarrow$ each effect separately and then summ

2) NON LIN approach
apply ϑ, x, φ effect at a time, SYSTEM on general config obtaining NON LIN. displacement ! linearize it later

$\Downarrow$

here we prefer (2):

general configuration



$$y_{G_1} = h_{G_1} = L_1 \sin(\vartheta)$$

displacement $|S_{G_1}| = L_1 \vartheta$

$$x_{G_2} = (L_0 + x)\cos(\vartheta)$$
$$y_{G_2} = (L_0 + x)\sin(\vartheta)$$
$$x_c = x_{G_2} + L_2\cos(\vartheta) = (L_0 + x + L_2)\cos(\vartheta)$$
$$y_c = (L_0 + x + L_2)\sin(\vartheta)$$

follow connection to find G_3

+ POSITIVE increasing x_{G_3}

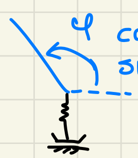
$$x_{G_3} = (L_0 + x + L_2)\cos(\vartheta) - L_3\cos(\varphi)$$

use (−) because $\cos(\varphi) < 0$ so to increase

(−) NEG, DECREASE y displacem.

$$y_{G_3} = (L_0 + x + L_2)\sin(\vartheta) - L_3\sin(\varphi)$$
$$y_D = (L_0 + x + L_2)\sin(\vartheta) - 2L_3\sin(\varphi)$$

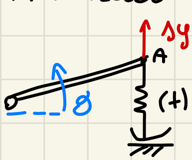
$$y_A = 2L_1\sin(\vartheta)$$



$\cos(\varphi) < 0$
$\sin(\varphi) > 0$

$\Delta l_1 = \Delta y_A$

k_1 connected vertically



$\Delta y_A > 0$ traction!

$\Delta y_A = y_A - y_{A_0}$

$\Delta l_2 = x$

same displacement of B

$\Delta l_3 = \Delta y_D = y_D - y_{D_0}$

$h_{G_1} = y_{G_1}$
$h_{G_2} = y_{G_2}$
$h_{G_3} = y_{G_3}$

same ref and sign convention

from here we should compute $\partial/\partial\vartheta$, $\partial/\partial x$, $\partial/\partial\varphi$ → NON LIN function

linearize to find [λ] LINEAR around static config.

computing as $\sin(\vartheta) \sim \sin(\vartheta_0)$

find so on $[K_g]$, $[K_{el_{II}}]$
for 3 bodies — for the 3 springs

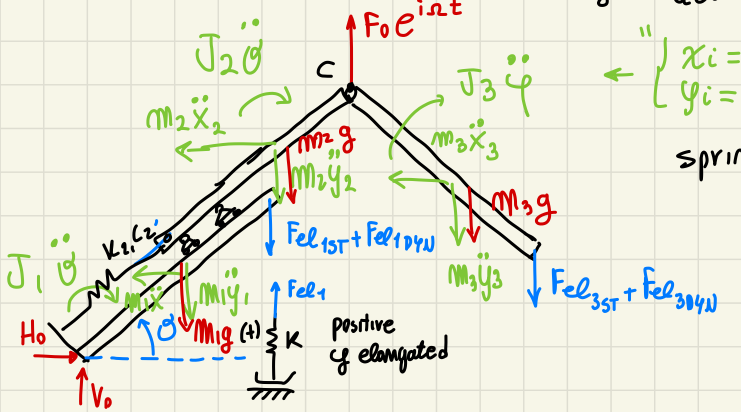
than on our system for shure we have angles $\neq 0$ → STATIC preload also so we have to compute II ORD

except for $\Delta L_2 = x$ LINEAR relation → $[K_{el_{II,2}}] = 0$

• QUESTION 4

compute FRF for Input $F_0 e^{i\Omega t}$ and output H_0 (constrain @ HINGE)

General config:



all forces acting on the OVERALL system

" $x_i = x_{G_i}$, $y_i = y_{G_i}$ "

spring 2 is internal

IT is easy to find H_0 by one computation! through

$$\sum_{SYST} F_x = 0 \Rightarrow H_0 - m_1\ddot{x}_{G_1} - m_2\ddot{x}_{G_2} - m_3\ddot{x}_{G_3} = 0$$

JUST the summ of inertia forces

with $\underbrace{\ddot{x}_{G_i}}_{1} = -\Omega^2 \underbrace{x_{G_i\,lin}}_{\text{linearized horiz. displacement}}$

because INPUT ARMONIC → OUTPUT ARMONIC

$x_{G_i} e^{i\Omega t} \to \frac{d^2}{dt^2} \to -\Omega^2 x_{G_i}$

check $[\lambda_m]$ to write $x_{G_{i\,LIN}}$ "LINEARIZED"

NOTICE! FRF is the linearized function! ratio I/O don't make sense to
use NONLIN. function

If you want compute V_0: from $\Sigma F_y = 0$

$$\underbrace{V_0 - (m_1+m_2+m_3)g}_{\text{static components of } V_{0\,ST}} - m_1\ddot{y}_{G_1} - m_2\ddot{y}_{G_2} - m_3\ddot{y}_{G_3} - F_{el_{ST_1}} - F_{el_1} - F_{el_{3ST}} - F_{el_3} + F_0 e^{i\Omega t} = 0$$

dynamic terms of V_0!

$F_0 e^{i\Omega t}$ INPUT $\rightarrow \underline{f}_{ph} = \{F_0\} e^{i\Omega t}$ $\quad \underline{f} = [\lambda_f]^T \underline{f}_{ph} = \underbrace{[\lambda_f]\cdot\{F_0\}}_{\underline{f}_0} e^{i\Omega t}$

$\underline{f}_0$ complex vector without time dependencies

from $[M]\ddot{\underline{x}} + [C]\dot{\underline{x}} + [K]\underline{x} = \underline{f}_0 e^{i\Omega t}$

$\rightarrow \underline{x} = \underline{x}_0 e^{i\Omega t}$

$$\underline{x}_0 = \left\{ \left(-\Omega^2[M] + i\Omega[C] + [K]\right)^{-1} \right\} \underline{f}_0$$

on MATLAB for loop all harmonic quantities

$\Downarrow$

$\theta = x(1) = \theta_0 e^{i\Omega t} = \left(|\theta_0| e^{i\psi_1}\right) \cancel{e^{i\Omega t}}$ we neglect that armonic part

NOT θ_{ST} @ eq.

$V_0 = m_1 \underbrace{(-\Omega^2 y_{G_1})}_{\text{just complex number}} + \dots + F_0 \cancel{e^{i\Omega t}}$

While for TIME HISTORY we assign a COMPLEX INPUT with given phase

$F_{01} = 100N$
$\psi_1 = \pi/6$ $\rightarrow$ so it's defined $F_{01} e^{i(\Omega t + \psi_1)} \Rightarrow F_0 = F_{01} e^{i\psi_1}$ without armonic part

$\Vert$

$F_{01}\exp(i\psi_1)$ MATLAB

complex force, having NONZERO ref and amplitude $F_{01} \neq 1$

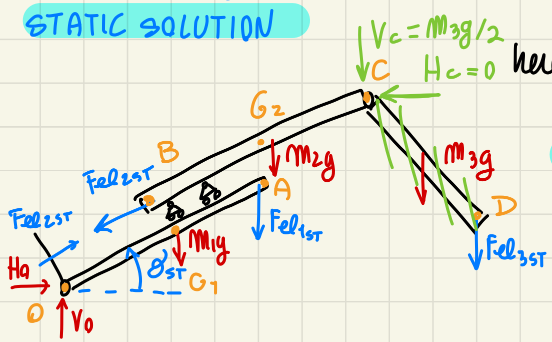
$\measuredangle F_0 \neq 0 \rightarrow$ initial phase

• QUESTION 5 computation of $\Delta l_{01}, \Delta l_{02}, \Delta l_{03}$

STATIC configuration: starting point of any dyn. config

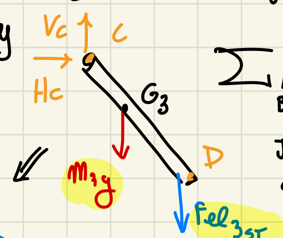
with changing θ_0, φ_0 respect the previous one! → changing Δl_0

STATIC SOLUTION



here I must compute Fel_{iST}!

EQ 1) considering JUST 3RD Rigid body



$$\sum M_c = 0$$
BODY3

JUST depends on:

$$Fel_{3ST}\, 2\cancel{L_3} \cos \cancel{\cos(\varphi_{ST})} + m_3 g \cancel{L_3} \cancel{\cos(\varphi_{ST})} = 0$$

$$Fel_{3ST} = -m_3 g/2 \Rightarrow \Delta L_{3ST} = Fel_{3ST}/K_3$$

half of the weight is supported by the spring!

then we can compute V_c, H_c useful ← later maybe by substituting BAR 3 JUST by V_3, H_3 acting on BAR 2

easily $H_c = 0$, $\sum F_x = 0$ No horizont forces

$$\sum F_y \Rightarrow V_c = m_3 g + Fel_{3ST} = \frac{m_3 g}{2}$$

(*)

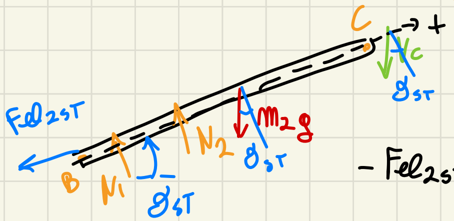
so we can remove BAR 3 on the system and put instead V_c on point C

EQ 2) to find $\Delta l_{01}, \Delta l_{02}$

so $Fel\ 1/2\, ST$ ...

We can consider an eq to delete H_0, V_0 → considering only 2^{ND} BAR we have a constrain acting vertically and moments due to the CARTS

$N_1, N_2 \perp Fel_{2ST}$ direction BC



$$\sum F_{//BC} = 0$$
BODY2

sum of forces along BC, eliminate N_1, N_2:

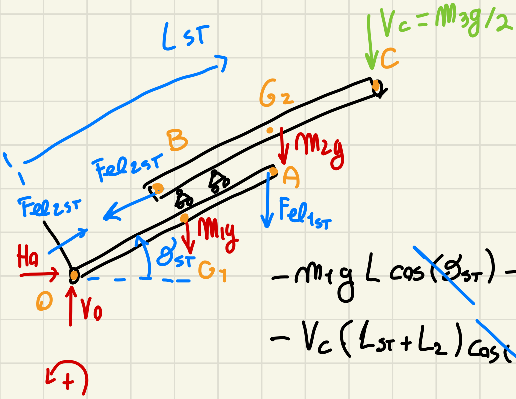
$$-Fel_{2ST} - m_2 g \sin\theta_{ST} - V_c \sin(\theta_{ST}) = 0$$

$$Fel_{2ST} = -(m_2 g + V_c)\sin(\theta_{ST}) = \underbrace{-\left(m_2 g + \frac{m_3 g}{2}\right)}_{<0}\sin(\theta_{ST}) \rightarrow \Delta l_{2ST} = \frac{Fel_{2ST}}{k_2}$$

compressed on static config ? YES

EQ 3) to find Fel_{1ST}

$\sum_{SUB-SYST} M_O = 0$ ← to remove the CONSTRAINTS acting on O on system of Body 1+2



$$-m_1 g L \cancel{\cos(\theta_{ST})} - Fel_{1ST}\, 2L_1 \cancel{\cos(\theta_{ST})} - m_2 g L_{ST} \cancel{\cos(\theta_{ST})}$$

$$- V_c(L_{ST}+L_2)\cancel{\cos(\theta_{ST})} = 0 \Rightarrow Fel_{1ST} \text{ from here}$$